

Chapter 7

Benefit/Cost Analysis Using AHP

In today's world, there are many decisions to be made at every level of management. Learning to closely examine all benefits, opportunities, costs and risk factors that will affect the decision to launch a project are important skills for decision-makers (Saaty 2012). One important problem faced in Benefit/Cost (B/C) analysis is the difficulty involved in assigning a value (monetary perhaps) to benefits, opportunities, costs and risks. The solution to this problem is to consider both tangible and intangible factors for the benefit/cost assessment process (Wijnmalen 2007).

7.1 AHP Benefit/Cost Analysis

Traditionally, B/C analysis has been performed to answer the following question: Which alternative will provide the greatest benefits with the lowest costs? Benefits are constituted by gains and pluses of all sorts-Economic, Physical, Psychological and Social. Similarly, costs are constituted by pains and losses of all sorts.

To perform an AHP B/C analysis, two separate hierarchies (one for benefits and one for costs) must be developed. For the benefits hierarchy, the criteria will be constituted by the expected benefits of the decision. Next, the criteria will be weighted using a pairwise comparison as usual. After this, the alternatives will be prioritized in terms of how beneficial they are. In other words, the comparison questions will always be: With respect to a given criterion, which is the best (most beneficial) alternative?

For the costs hierarchy, the criteria will be constituted by the expected costs of the decision. However, and this is an important consideration, the alternatives are prioritized in terms of how costly they are (i.e. a higher priority reflects a higher cost). For this purpose, the comparison question is: With respect to a given criterion, which alternative is more costly?

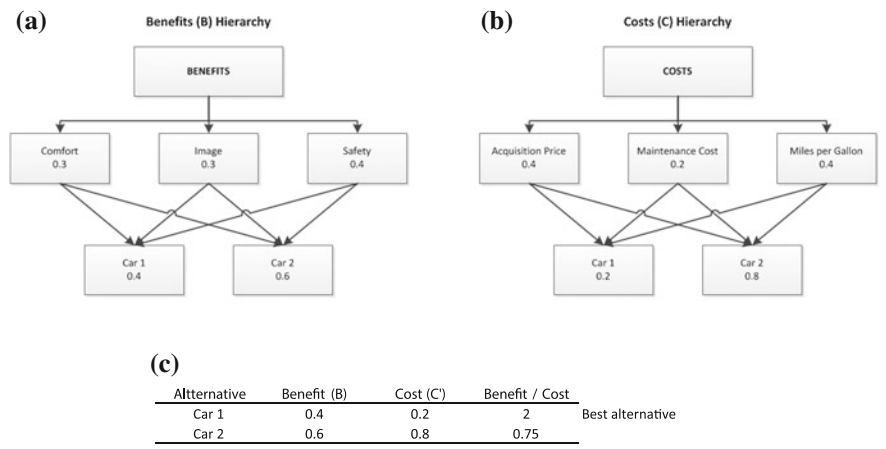


Fig. 7.1 AHP B/C analysis. **a** The higher the priority of the alternative, the more beneficial (or preferable). **b** The higher priority of the alternative, the more costly (or worse) alternative. **c** The higher B/C ratio, the more preferable the alternative

Finally, the ratio benefit/cost for the priority of each alternative is computed by dividing the benefit priority by the cost priority of each of the alternatives. The best alternative is the one that has the highest ratio.

To illustrate this with an example, let’s assume that we have decided to analyze the decision to purchase a car using a B/C analysis. The steps are as follows:

- Step 1: Build a Benefits Hierarchy, weight the criteria using pairwise comparison, and next prioritize the alternatives based on how beneficial they are (i.e. higher priority indicates higher benefit). Results for the example are shown in Fig. 7.1a.
- Step 2: Build a Cost Hierarchy, weight the criteria using pairwise comparison, and next prioritize the alternatives based on how costly they are (i.e. higher priority indicates higher cost). Results for the example are shown in Fig. 7.1b.
- Step 3: Calculate the B/C ratio for each alternative. The alternative with the highest B/C ratio will be the best. Results for the examples are shown in Fig. 7.1c.

A more detailed explanation of each example follows.

Step 1: Build a Benefits Hierarchy

Compare the criteria pairwise to obtain the benefits criteria weights (e.g. “With respect to the benefit expected when buying a car, which criterion is more important: Comfort or Image?” and so on). This procedure is the same as we have done before to determine the relative importance of criteria. As shown in Fig. 7.1a, Safety is the most important criterion (weight: 0.4), followed by Comfort and Image (weight: 0.3 in both cases).

Next, compare the alternatives pairwise with respect to each criterion (i.e. “With respect to Comfort, which alternative is more beneficial: Car 1 or Car 2?” “With respect to Image, which alternative is more beneficial: Car 1 or Car 2?” “With respect to Safety, which alternative is more beneficial: Car 1 or Car 2?”). Finally, calculate the overall priorities.

The Benefits hierarchy in Fig. 7.1a shows the final results. According to this we should purchase the Car 2 which is the most preferable (priority: 0.6) in terms of benefits, with respect to Car 1 (priority: 0.4). However, this would be the right decision if we were only to consider the benefits; but we still need to consider the costs.

Step 2: Build a Cost Hierarchy

Compare the criteria pairwise to obtain the cost criteria weights (e.g. “With respect to the cost of buying a car, which criterion is more important: Acquisition Cost or Maintenance Cost?” and so on). This procedure is the same as we have done before to determine the relative importance of criteria. As shown in Fig. 7.1b Maintenance Cost (weight: 0.2) is the least important cost criterion while Acquisition Price and Miles per Gallon consumption are much more important (weight: 0.4 each).

Next, compare the alternatives pairwise with respect to each criterion (i.e. “With respect to Acquisition Price, which alternative is more costly: Car 1 or Car 2?” “With respect to Maintenance Cost, which alternative is more costly: Car 1 or Car 2?” “With respect to Miles per Gallon consumption, which alternative is more costly: Car 1 or Car 2?”). Finally, obtain the overall priorities. The Cost Hierarchy in Fig. 7.1b shows the final results. According to this we should purchase the Car 1 which is the least costly (priority: 0.2) with respect to Car 2 (priority: 0.8). However, this would be the right decision if we were only to consider the costs and naturally, wanted to purchase the least costly vehicle. However, in B/C analysis we need to integrate both benefits and costs as shown next.

Step 3: Calculate the B/C Ratio for each Alternative

To find out which is the best solution, simply divide the benefit by the cost (B/C) for each alternative as shown in the last rightmost column in Fig. 7.1c. The highest B/C ratio is the best alternative, taking into account both benefits and costs. In this case, “Car 1” constitutes the best alternative when taking into account both benefits and costs.

7.2 AHP Benefit*Opportunity/Cost*Risk (BO/CR) Analysis

Traditionally, there are two ways to perform a B/C analysis involving monetary situations: First, it is possible to calculate the ratio Benefit (B) to Costs (C) as B/C. This is called B/C ratio approach. If the benefits are greater than the costs (i.e. ratio $(B/C) > 1$), the project will be profitable. If there are several possible alternatives to

choose among, they will be usually prioritized in terms of their B/C ratios from highest to lowest. This is the approach we have used in our previous example.

A second approach is possible based on the net value of the proposed project; that is, if the difference $(B - C)$ is positive (i.e., $(B - C) > 0$), this net value will constitute the net benefit of the project. On the other hand, if this net value is negative, the project will derive losses. Again, if there are several possible alternatives to choose among, they will be prioritized in terms of their net value from highest to lowest.

AHP has extended B/C analysis by allowing mixing tangible (e.g. monetary values) and intangible (e.g. image) considerations. In other words, it is possible to consider various tangible and intangible factors to evaluate the merits, in terms of benefits and costs, of the alternatives (Saaty 2012).

Also, AHP allows considering opportunities (O) and risks (R) as part of the B/C analysis in what is called BO/CR analysis (Saaty and Ozdemir 2004). An opportunity is defined as a potential (but not certain) benefit or gain while a risk is defined as a potential (but not certain) cost or loss.

The original proposed formulas that could be considered an extension of the original (B/C) and $(B - C)$ analyses are¹:

Multiplicative Ratio:

$$(B * O) / (C * R)$$

In this expression, it is intuitively noticeable that alternatives with larger numerators (either because B or O are large) and/or smaller denominators (either because C or R are small) will be the most attractive alternatives overall because it will give you the highest ratio. This basically constitutes an extension of the B/C rationale.

Additive with Subtraction:

$$(B + O) - (C + R)$$

The additive with subtraction analysis is an extension of the original net value approach $(B - C)$. This time there are potential benefits (called opportunities) as well as potential costs (called risks). Therefore, the opportunities add to the benefits as well as the risks add to the costs.

In AHP BOCR analysis is performed as follows: first, a hierarchy is built separately for Benefits (B), Opportunities (O), Costs (C) and Risks (R); second, the

¹For completeness purpose, we should indicate that another proposed formula by Saaty (2012) to integrate BO/CR is the additive/reciprocal: $B + O + 1/C + 1/R$. However, the use of reciprocals has been strongly questioned by Millet and Schoner (2005), who have argued that taking cost and risk reciprocals actually distorts their originally common scale, thereby producing confounding results. In any case, this formula is rarely used in practice and for this reason is not further discussed in this book.

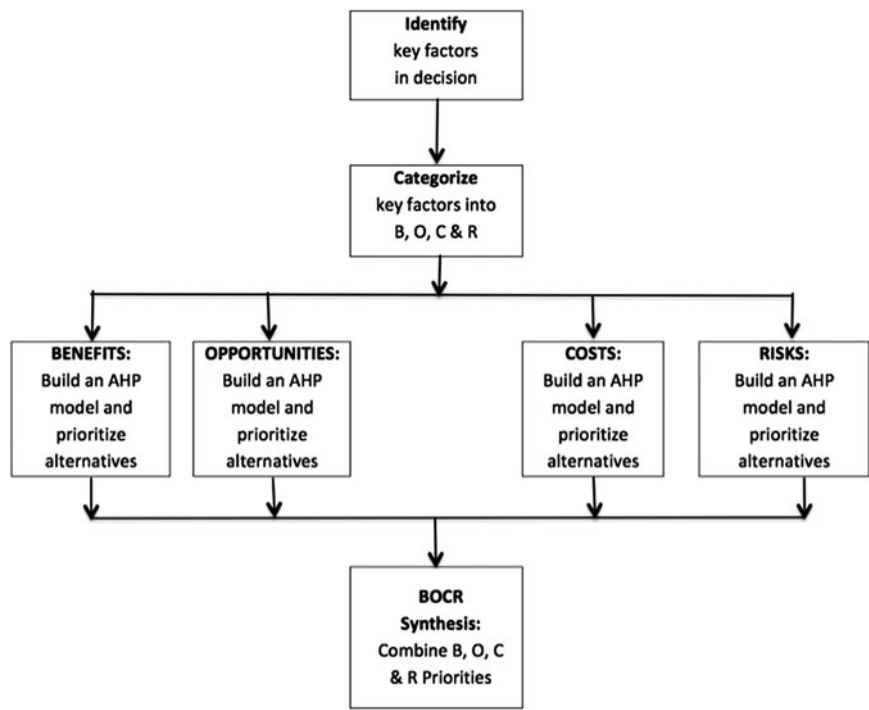


Fig. 7.2 Overview of the BOCR analysis process

alternatives are prioritized with respect to B, O, C and R in their respective hierarchies and finally, they are combined using either the multiplicative or additive/subtractive formula shown previously. An overview of the process to conduct a BOCR analysis is given in Fig. 7.2.

The multiplicative ratio approach $(B * O)/(C * R)$ can be considered an extension of the well-known B/C ratio analysis previously used. In BO/CR analysis, opportunities refer to potential (but not certain) benefits while risks refer to potential (but not certain) costs as previously indicated. In other words, if a gain is possible but not certain, it constitutes an opportunity, while if a loss is possible but not certain it constitutes a risk. A simple rule of thumb is typically to ask the question: “Can we be reasonably sure this benefit will occur?” If the answer is ‘Yes’ it is a benefit, if the answer is ‘No’ it is an opportunity. A similar question is made to distinguish costs from risks.

Keep in mind that the priorities of the alternatives in the benefits and opportunities hierarchies reflect how preferable the alternative is (i.e. the higher the priority, the more desirable the alternative is in terms of benefits or opportunities) while the priorities of the alternatives in the costs and risk hierarchies reflect how undesirable the alternative is (i.e. the higher the priority, the more costly or riskier the alternative is). For this purpose, when evaluating the alternatives pairwise with respect

to the benefits (or opportunities), the comparison question should be: With respect to this given benefit (or opportunity), which is a better alternative? On the other hand, when evaluating the alternatives pairwise with respect to the costs (or risks), the comparison question should be: With respect to this given criterion, which alternative is more costly (or riskier)?

GOAL							
Assess University-Uganda University Initiatives from the Perspective of University's School of Education							
CRITERIA	Benefits	Opportunities	Costs	Risks			
SUB CRITERIA	Learning Potential 0.163	Interaction w/ Uganda Business 0.088	Training 0.280	Terrorism 0.493			
	Mission Sync 0.540	Social Responsibility 0.195	Resources 0.627	Sickness & Disease 0.311			
	Recognition 0.297	Project Planning 0.717	Building Renovation & Maintenance 0.094	Accidents & Disasters 0.196			
ALTERNATIVES	Bright Kids Uganda	Entebbe School	Children's Rights	End Human Trafficking			

	Benefits	Opportunities	Costs	Risks	B/C	B•O/C•R	Ranking
Bright Kids	0.263	0.185	0.334	0.297	1.42	0.885	3
Entebbe School	0.550	0.447	0.458	0.167	1.23	3.297	1
Children's Rights	0.077	0.140	0.096	0.480	0.55	0.161	4
Human Trafficking	0.108	0.225	0.112	0.054	0.48	2.000	2

Fig. 7.3 BO/CR analysis for international opportunities

7.3 A BO/CR Analysis Example

Mu (2016) reports a BO/CR analysis of four proposed university initiatives to collaborate with institutions in Uganda: Bright Kids Uganda, Entebbe School, Children's Rights and End Human Trafficking. A team of university decision-makers performed a BO/CR analysis from the perspective of the School of Education (Fig. 7.3, upper half). They created a hierarchy consisting of 4 criteria: Benefits, Opportunities, Costs and Risks and considered the important decisional factors under each of the criteria. After prioritizing these factors via pairwise comparison they calculated the local priorities of the alternatives and obtained the overall priorities. As can be seen in Fig. 7.3 (lower half), the Entebbe School initiative was the most convenient in terms of Benefits (Priority: 0.550) and Opportunities (Priority: 0.447). However, this alternative was also the most Costly (Priority: 0.458) but the least Risky (Priority: 0.167). After performing a $B * O/C * R$ calculation of the priorities of the alternatives, it can be seen that the best alternative, taking into account benefits, opportunities, costs and risks is given by Entebbe School which has a BO/CR multiplicative ratio of 3.297, much higher than the other alternatives.

7.4 Conclusion

While BO/CR analysis is very useful for the evaluation of economic alternatives you must keep in mind some final comments for this type of analysis:

- Do not forget that the comparison questions for alternatives in the cost (risk) hierarchies are the opposite of the questions in the benefits (opportunities) hierarchies. In the benefits (opportunities) hierarchy you select the best (more beneficial or opportunistic) alternative with respect to each criterion but in the cost (riskier) hierarchy you select the worst (more costly) alternative with respect to each criterion.
- Not all the problems can be (or must be) converted to B/C or BO/CR analysis but there are some that are particularly suitable for this type of analysis (e.g. economic decisions).
- Not all the 4 hierarchies are needed. There may be problems suitable for B/C Analysis (B/C ratio), B/CR analysis ($B/C * R$) or similar ones.
- When using *Super Decisions* or any other software, it may be easier to create and save different AHP models for each of the B, C, O and R hierarchies and make the corresponding BO/CR ratio calculations outside the software, in a spreadsheet, as shown in our examples.

References

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